Reg. No. | | | | | | | | | | |

[20ME31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

MATERIAL SCIENCE AND ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	With a neat sketch, explain interstitial imperfection, substitutional imperfection and edge dislocation.	07
	b.	Define APF. Determine the APF value for BCC and HCP crystal structures.	07
2.	a.	What is solidification and explain the two stages of solidification in pure metals.	08
	b.	Define Creep and with neat sketch explain the stages of creep.	06
3.	a.	Draw a neat sketch of Iron Carbon equilibrium diagram, label all the phases and explain them.	10
	b.	Discuss the any two types of corrosion in metals.	04
		OR	
4.	a.	With a phase diagram explain the peritectic reaction	10
	b.	Discuss about the GIBB'S phase rule	04
5.	a.	Discuss the applications of high carbon steel and aluminum.	04
	b.	Define hardenability and explain Jominy end quench test.	10
		OR	
6.	a.	Explain austempering heat treatment process.	06
	b.	Explain the Pack carburizing and mention its advantages and disadvantages.	08
7.	a.	Define composites, Explain the classification of the composites based on reinforcement.	10
	b.	Briefly discuss about the characterization of Alumina and SiC.	04
		OR	
8.	a.	Explain the role of matrix in the composites. Also brief the applications of MMCs in automobile applications.	07
	b.	With neat sketch, explain sandwich composite and laminar Composites.	07